

ANSWER KEY — Parent / Teacher Copy**Chemistry — Unit 5: Molecular Shape**

Quiz [B] — Bonding Sites, Lone Pairs, Shape Names, and 3-D Drawing

Section 1 — Identify Bonding Sites and Lone Pairs (5.1.1)**1. CO₂**

(4 pts — 2 pts each)

- a. Bonding sites on C: **2**
- b. Lone pairs on C: **0**

Why:

CO₂ is O=C=O. Even though there are two *double bonds*, each double bond still counts as ONE bonding site. C has 2 bonding sites and uses all 4 of its valence electrons in the two double bonds—zero left for lone pairs. Result: linear shape, 180° bond angle.

Common mistake to watch for: Students may count 4 bonding sites by counting each line in O=C=O. Remind Kai: it's the *number of bonded atoms* attached, not the number of bond lines.

2. PCl₃

(4 pts — 2 pts each)

- a. Bonding sites on P: **3**
- b. Lone pairs on P: **1**

Why:

Phosphorus has 5 valence electrons. It forms 3 bonds with the 3 Cl atoms (using 3 electrons). The remaining 2 electrons form 1 lone pair. 3 bonds + 1 lone pair = 4 electron pairs total = domain geometry is tetrahedral, molecular shape is *trigonal pyramidal*. PCl₃ is structurally identical to NH₃—same shape, different atoms.

3. NH₃ vs. NF₃

(6 pts — 1 pt per blank + 2 pts explanation)

- a. NH₃: bonding sites = **3** lone pairs = **1** shape = **Trigonal Pyramidal**
- b. NF₃: bonding sites = **3** lone pairs = **1** shape = **Trigonal Pyramidal**
- c. **Yes, they have the same shape. Molecular shape depends only on the number of bonding sites and lone pairs on the central atom. Both NH₃ and NF₃ have N with 3 bonding sites and 1 lone pair, so both are trigonal pyramidal. It doesn't matter whether the bonded atoms are H or F.**

Key insight for Marcus:

This is a powerful concept. The shape of a molecule is determined entirely by what's on the *central atom* (bonds + lone pairs). The identity of the outer atoms changes bond polarity (covered in 5.1.4) but not the geometric shape.

Section 2 — Name Molecular Shapes (5.1.2)**4. Shape table**

(6 pts — 2 pts per row: 1 pt name + 1 pt angle)

# Bonding Sites	# Lone Pairs	Shape Name	Bond Angle
3	0	Trigonal Planar	120°
2	1	Trigonal Planar Bent	<120°
3	1	Trigonal Pyramidal	~107°

Distinction Marcus should know: Trigonal Planar Bent vs. Tetrahedral Bent

Both are called “bent,” but they come from different domain geometries. *Trigonal planar bent* (2 bonds, 1 lone pair) has a domain geometry of trigonal planar—all 3 electron pairs in one plane, bond angle <120°. Example: SO₂. *Tetrahedral bent* (2 bonds, 2 lone pairs) has a domain geometry of tetrahedral—bond angle ~104.5°.

Example: H₂O. Kai may confuse these two—the key is counting lone pairs.

5. H₂S vs. H₂O

(6 pts)

Full credit answer:

Yes, the student is correct about the *shape*. Both H₂S and H₂O have a central atom with 2 bonding sites and 2 lone pairs, so both are tetrahedral bent (bent). The VSEPR table predicts a bond angle of ~104.5° for this category (based on H₂O). Accept 104.5° as the answer at this level. Note for Marcus: the actual measured angle of H₂S is closer to 92° because sulfur is a larger period-3 atom and uses unhybridized p-orbitals, which are naturally 90°. H₂O follows the VSEPR table closely; H₂S deviates significantly. Kai does not need to know the 92° value for this standard.

Scoring guide:

2 pts — States student is correct; 2 pts — Names shape (bent / tetrahedral bent); 2 pts — Explains lone pair compression effect on bond angle

6. Mystery molecule (2 bonds, 2 lone pairs)

(6 pts — 1.5 pts each)

- Domain geometry: **Tetrahedral**
- Molecular shape: **Tetrahedral Bent (Bent)**
- Bond angle: **104.5°**
- Example molecule: **H_2O** (*also accept H_2S*)

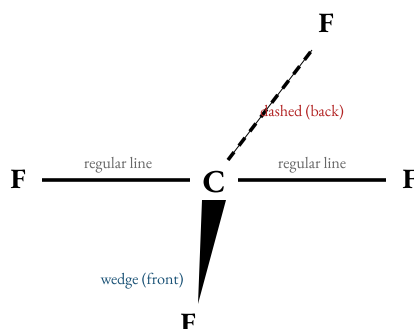
Domain geometry vs. molecular shape explained:

Domain geometry = the arrangement of ALL electron pairs (bonds + lone pairs). Molecular shape = the arrangement of the ATOMS only. For H_2O : 4 electron pairs $\rightarrow$ domain is tetrahedral. But only 2 bonds exist between atoms, so the shape seen is bent. This two-name system is tested constantly in Unit 5.

Section 3 — Draw 3-D Structures (5.1.3)**7. CF_4 — Tetrahedral 3-D Drawing**

(8 pts)

Same structure as CH_4 —F replaces H at each bond end. Tetrahedral: 4 bonds, 0 lone pairs, 109.5° .

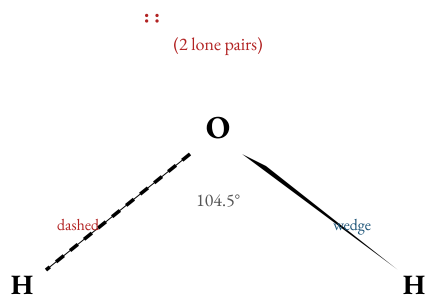
**Scoring rubric (8 pts):**

- 1 pt — Wedge bond (solid filled triangle) present and pointing toward viewer
- 1 pt — Dashed bond (dashed line) present and pointing away from viewer
- 2 pts — Two regular bonds in the plane of the paper
- 4 pts — F atom at each of the 4 bond ends (1 pt each)

Accept any valid tetrahedral orientation. The shape of CF_4 is identical to CH_4 —this problem tests whether Kai understands that atom identity doesn't change geometry.

8. H_2O — Tetrahedral Bent 3-D Drawing

(10 pts)



Sample explanation (full credit):

“H₂O has a bond angle of 104.5° rather than 109.5° because it has 2 lone pairs on oxygen, while CH₄ has 0 lone pairs. Lone pairs repel more strongly than bonding pairs, pushing the 2 H atoms closer together and compressing the H–O–H angle below the tetrahedral ideal of 109.5°.”

Scoring rubric (10 pts):

- 3 pts — Drawing shows bent shape (O in center, 2 H bonds angled downward)
- 3 pts — Bond types labeled (wedge + dash, or wedge + regular, or dash + regular)
- 2 pts — Both lone pairs shown/labeled above O
- 2 pts — Explanation references lone pairs causing angle compression below 109.5°

Total: _____ / 50